

K Shiv Kumar

(☎) +91-8305014305

(✉) kumar[dot]shiv301[at]gmail[dot]com - (in) linkedin.com/in/kumarkshiv - (github) github.com/kumarkshiv

“Engineer with **more than 1 year** of experience in industry, and **more than 4 years** of research experience in programmable data plane (P4 and eBPF) and compiler framework (LLVM). Skilled in C++, Python, and systems programming, with expertise in program analysis of low-level eBPF instructions and experimentation. With experience in academic research, I’m eager to deepen my expertise in designing scalable infrastructure with a long-term focus on working with large-scale data centers for AI workloads.”

EDUCATION

IIT Hyderabad - MS(CSE)

Kandi, Telangana

Completed 4 years of PhD research ([NETX Lab](#)); degree conferred as **MS** - (GPA: 7.60/10)

2021–2026

Research: eBPF, P4 and Compilers

IIIT Naya Raipur

Naya Raipur, Chhattisgarh

MTech (Computer Science and Engineering) - (GPA: 8.78/10)

Jul 2019–Jul 2021

Shri Shankaracharya Engineering College

Bhilai, Chhattisgarh

B.E. (Information Technology and Engineering) - (Percentage: 74.48/100)

Mar 2011–Mar 2015

TECHNICAL SKILLS

Core: XDP/eBPF, TC/eBPF, P4, C++, Python

Libraries and Tools: Libbpf, bpfman, Mininet, Docker, Kubernetes, Github, VirtualBox, Vagrant, SQL

Frameworks: BMv2, Tofino, Barefoot Runtime Interface (BRI), LLVM

Networking Concepts: Networking fundamentals, Standard network protocols, TCP/IP stack, Linux Networking stack, SDN, Programmable networks, Host networking stack

Other Skills: Customization with apex classes and visualforce pages, code migration using tools like Workbench, Jenkins, Eclipse and Git, data migration using Apex dataloader.

PROJECTS AND RESEARCH PUBLICATIONS

“In-Network Probabilistic Monitoring Primitives under the Influence of Adversarial Network Inputs” | APNET '23: (🔗)

- Analyzed vulnerabilities of monitoring primitives in programmable data planes against adversarial inputs.
- Demonstrated FlowRadar pollution attacks where a few malicious flows corrupted telemetry data.
- Quantified attack impact, revealing a 99% drop in FlowRadar’s accuracy under targeted adversarial conditions.
- **Tools & Frameworks:** Python, NetworkX, Wireshark.

“DBVal: Validating P4 Data Plane Runtime Behavior” | SOSR '21: (🔗)

- Developed DBVal, a runtime validation system to detect packet-processing errors in P4 data planes.
- Implemented assertion-based tracking of tables and actions, enabling line-rate validation of runtime behavior.
- **Tools & Frameworks:** Python, P4 language, BMv2, Mininet, Wireshark.

“Understanding eBPF-based Network Function Behavior from its Bytecode” | Under Submission

- Developed a system to validate the correctness and interactions of third-party eBPF network function bytecode.
- Applied program analysis to extract functional properties of eBPF network functions from the bytecode.
- **Tools & Frameworks:** C++, STL library, LLVM, eBPF/XDP, Libbpf, Linux kernel, bpftool, bpfman.

PROFESSIONAL EXPERIENCE

Application Development Analyst (*Accenture, Pune, India*)

Mar 2016–Dec 2017

Client: Splunk

- Developed various enhancements based on configuration and customization as per the Client’s requirements.
- Successfully integrated Docebo Learning Management System into Splunk’s Salesforce environment.

Client: Prudential

- Delivered various requirements based on various customizations and configurations in Salesforce.

Client: Royal Bank of Canada Insurance

- Delivered various requirements based on apex visualforce, configurations and outlook integration.

CERTIFICATIONS

- **Intel Connectivity Academy Level 1** | Issued by: Intel ([🔗](#))
 - Covered data/control plane development with training in advanced parsing, externs in TNA (Tofino Native Architecture), Barefoot Runtime Interface, and Tofino ASIC architecture.

AWARDS AND HONORS

- ACM SIGMETRICS Student grant: For presenting our ongoing research on “*Detecting Adversarial Attacks on Bloom Filters in P4 Data Plane Systems*” (2022). ([🔗](#))
- COMSNETS Travel Grant: For attending the Conference (2022). ([🔗](#))
- Apex Award - Extra Miller, in Accenture India: Recognized for consistently achieving the client objectives (2017).

TALKS AND PRESENTATIONS

MS, IIT Hyderabad

2021–2025

- Attended first eBPF Day India workshop organized by IISc and eBPF Foundation (2024). ([🔗](#))
- Conducted a session titled “*Introduction to eBPF*” in a SERB-sponsored Workshop on “SDN: Software Defined Networking Architectures and Applications” at IIIT Kottayam (2023). ([🔗](#))
- Delivered a seminar talk on “*Introduction to eBPF*” in CSE Department seminar series at IIT Hyderabad (2023)
- Presented the work “*Detecting Adversarial Attacks on Bloom Filters in P4 Data Plane Systems*” at the Graduate Forum, PerfNA Workshop, IIT Bombay (2022). ([🔗](#))

VOLUNTEERING AND ASSISTANTSHIPS

MS, IIT Hyderabad

2021–2025

- Teaching Assistant for courses *Computer Networks*, *Advanced Computer Networks* and *Topics in Networks*.
- Volunteered in the *International Conference on Distributed Computing and Networking*. ([🔗](#))
- Volunteered in *ACM Workshop on Research Opportunities in Computer Science*. ([🔗](#))

MTech, IIIT Naya Raipur

2020–2021

- Teaching Assistant for courses *Advanced Computer Networks* and *Software Defined Networks*.

ADDITIONAL INFORMATION

- **Extracurriculars:** Completed 5 KM Run, organized by Heartfulness and FIT India (2023). ([🔗](#))
- **Hobbies:** Reading books, Playing chess, Doing yoga, Running.

REFERENCES

Supervisor: Dr. Praveen Tamma ([🔗](#))

Assistant professor,
Dept. of CSE,
IIT Hyderabad

Co-Supervisor: Dr. Ramakrishna Upadrasta ([🔗](#))

Associate Professor,
Dept. of CSE,
IIT Hyderabad